

Threshold-Vector Rigidity of a Frozen Subgroup Poset

Anonymous

Abstract

For each of the $2^{16} = 65536$ retained supports in a frozen named model, we collect the twenty subgroup-containment repair thresholds into one vector. Exactly twenty vectors occur, and a vector depends only on the generated subgroup. Its zero coordinates are precisely the principal ideal below that subgroup, so the vector recovers the closure. Moreover, subgroup inclusion is equivalent to reversed coordinatewise order between the twenty closure vectors. The exact fibre spectrum is $\{32:6, 64:4, 96:4, 192:2, 1760:2, 30400:2\}$. Independent point-set and antichain reconstruction, a finite-lattice symbolic check, clean replay, and twenty-three hostile mutations certify the result. The scope is finite and obeys `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Finite setting

Let $L = \{S_1, \dots, S_{16}\}$ be the frozen named support and let $\Phi(A) \leq Q = \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$ be the subgroup generated by a retained support $A \subseteq L$. Write $H_0, \dots, H_{19}$ for the twenty actual subgroups in the source-bound C75 order. For $D = L \setminus A$, C80 defines

$$\tau_H(D) = \min\{|R| : R \subseteq D, H \subseteq \Phi(A \cup R)\}.$$

We study the complete threshold vector

$$v(A) = (\tau_{H_0}(D), \dots, \tau_{H_{19}}(D)) \in \mathbb{Z}_{\geq 0}^{20}.$$

This construction uses the actual point-set subgroups. The C75 lifted ambient group has order 11520, whereas the effective C76 action on the sixteen labels has order 1920; these objects remain distinct throughout.

2 Rigidity theorem

Theorem 1 (Threshold-vector subgroup-poset rigidity). *For every $A \subseteq L$ there is a vector $w_{\Phi(A)}$ such that $v(A) = w_{\Phi(A)}$. Exactly twenty distinct vectors occur, one for each actual subgroup of Q , and*

$$\{i : v(A)_i = 0\} = \{i : H_i \leq \Phi(A)\}.$$

Consequently the zero coordinates recover $\Phi(A)$. For actual subgroups $K_0, K_1 \leq Q$,

$$K_0 \leq K_1 \iff w_{K_0} \geq w_{K_1} \text{ coordinatewise.}$$

Proof. The threshold definition gives $\tau_H(D) = 0$ exactly when no restoration is needed, equivalently $H \leq \Phi(A)$. Thus the zero set is the principal ideal below $\Phi(A)$ and recovers its unique maximum.

The finite certificate enumerates all 65536 supports and verifies that the C80 vector is constant on each point-set closure fibre. All twenty C75 subgroups occur as closures, and their vectors have distinct zero ideals; hence exactly twenty vector classes occur. The certificate also checks the forward coordinatewise inequality for all 20^2 ordered subgroup pairs. For the converse, $w_{K_0} \geq w_{K_1}$ implies that every zero coordinate of w_{K_0} is zero in w_{K_1} . Therefore $\downarrow K_0 \subseteq \downarrow K_1$, which implies $K_0 \leq K_1$. $\square$

3 Exact fibres and finite poset

The closure/vector fibre inventory is

fibre size	32	64	96	192	1760	30400
number of subgroups	6	4	4	2	2	2

and the weighted sum is $6(32) + 4(64) + 4(96) + 2(192) + 2(1760) + 2(30400) = 65536$. The recovered subgroup poset has forty cover relations and 102 comparable ordered pairs when its twenty reflexive pairs are included.

The producer rebuilds every closure from the 54-point addition law and groups the frozen C80 profile rows. The independent checker takes a different route: it enumerates all twenty subgroups, finds every inclusion-minimal named support whose closure contains each target, and evaluates

$$\tau_H(L \setminus A) = \min_{M \in \mathcal{M}_H} |M \setminus A|$$

for all $20 \cdot 65536$ entries. A SymPy check verifies that the inclusion zeta matrix has determinant one, that its inverse is the integer Möbius matrix, and that all 400 subgroup meet/join pairs remain in the finite list.

4 Audit and scope

The producer and checker raw-byte bind the C75, C76, and C80 evidence and manifests. The canonical C85 evidence SHA-256 is `22bdaf9fa2fe08532b45eae51cf7704a1509764b5a09f10eebb98012224be152`. Clean replay preserves this digest, and all 23/23 hostile semantic mutations are rejected.

The theorem concerns only a finite named-support closure and its twenty actual subgroups. It does not assert a full Burnside ring or table of marks, arithmetic or local data, Euler factors, root numbers, automorphy, or a Hilbert–Polya operator.